



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Course: BVWS103-OWASP Top 10 Vulnerabilities And
Exploitation Techniques

Student Name: Abubakari-Sadique Hamidu
Student ID: 2025/FWSD/11392

Instructor: Aminu Iddris (CCNA, CompTIA, CEH, OSCP, CISSP, CISM)

Lab 12: Insufficient Logging and Monitoring – Advanced Detection
Techniques

Lab 12: Insufficient Logging and Monitoring – Advanced Detection Techniques

1. Objective

The objective of this lab was to enhance logging mechanisms in a PHP login system, analyze logs for suspicious activities, and implement a basic real-time alert mechanism to detect potential security incidents such as brute-force attacks.

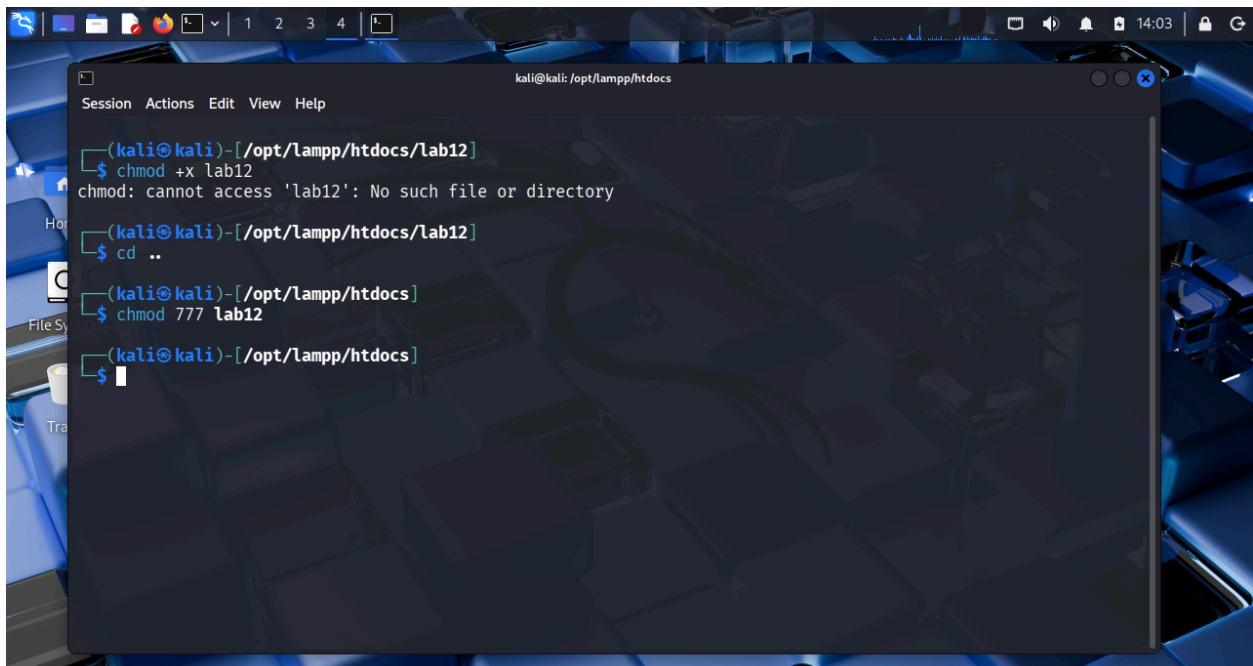
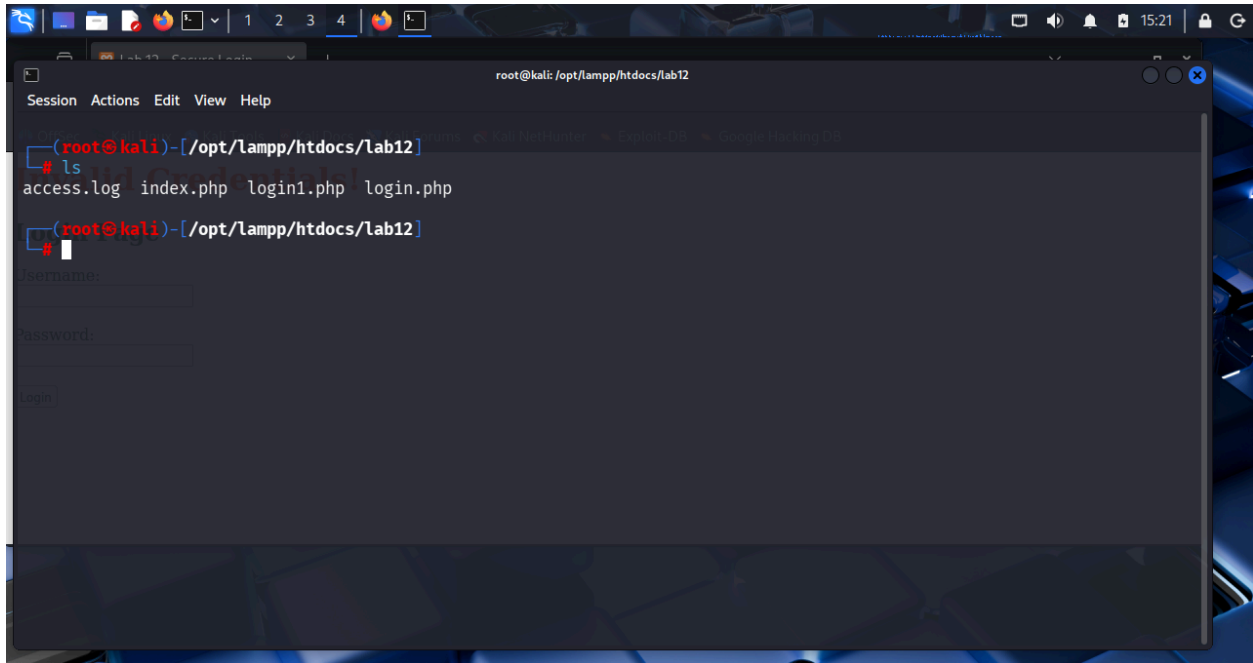
2. Implementation

2.1 Enhanced Logging

The login.php file was modified to log detailed information for every login attempt, including:

- Timestamp
- Username entered
- IP address
- User-Agent
- Login status (SUCCESS or FAILURE)

All login attempts are recorded in access.log.



```
root@kali: /opt/lampp/htdocs
GNU nano 8.7 login.php *
<?php
session_start();
$log_file = __DIR__ . '/access.log'; // __DIR__ returns the current directory

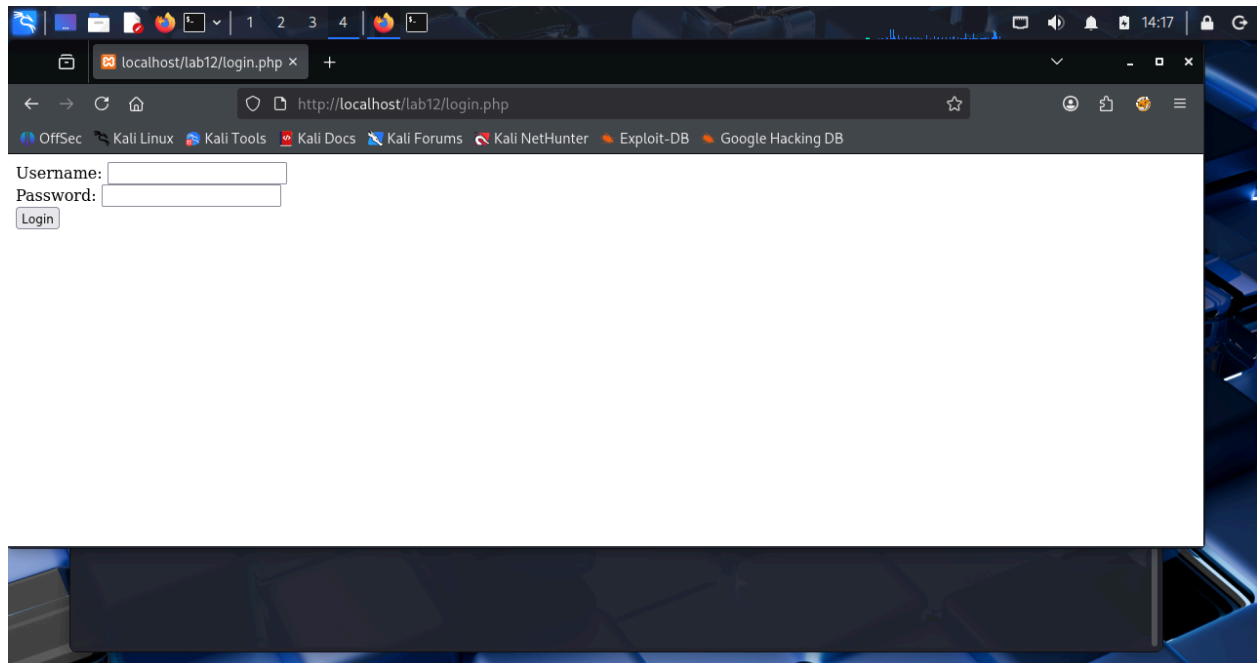
// Hardcoded credentials
$username = "admin";
$password = "password123";

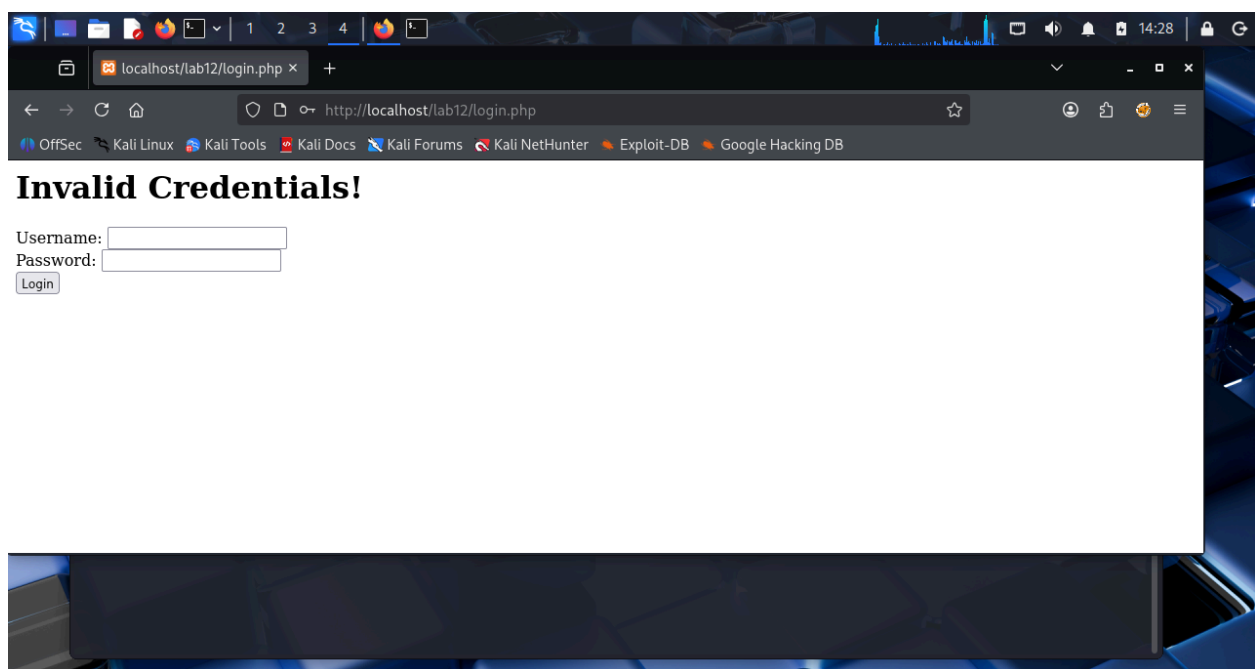
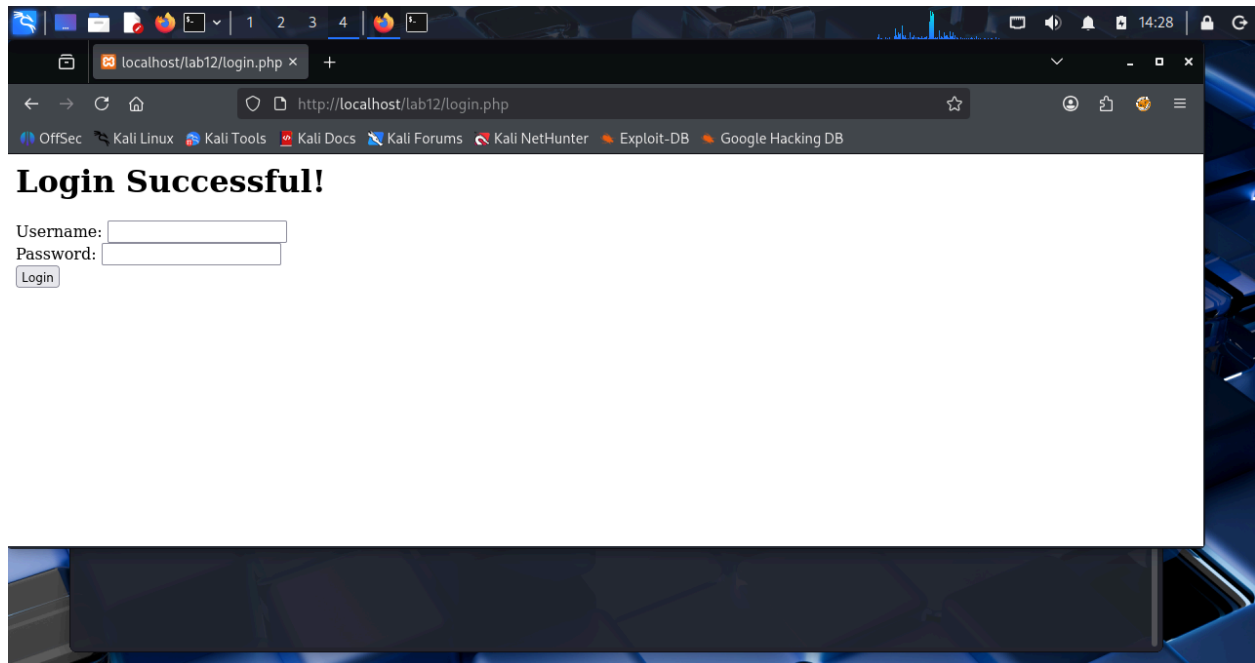
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $input_user = $_POST['username'];
    $input_pass = $_POST['password'];
    $ip_address = $_SERVER['REMOTE_ADDR'];
    $user_agent = $_SERVER['HTTP_USER_AGENT'];

    // Log all login attempts (both success and failure)
    $status = ($input_user === $username && $input_pass === $password) ? 'SUCCESS' : 'FAILURE';
    $log_message = "[" . date("Y-m-d H:i:s") . "] Login attempt by $input_user from IP: $ip_address, User-Agent: $user_agent";
    file_put_contents($log_file, $log_message, FILE_APPEND);

    if ($status === 'SUCCESS') {
        $_SESSION['user'] = $input_user;
    }
}
```

```
root@kali: /opt/lampp/htdocs/lab12
GNU nano 8.7 login.php *
}
<?php
<form method="POST">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required><br>
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br>
    <button type="submit">Login</button>
</form>
```





2.2 Log Verification

After performing multiple login attempts (both successful and failed), the access.log file was reviewed using:

```
cat access.log
```

The log entries showed detailed records such as:

- Failed login attempts (e.g., users: sadi, him, hero, admin, user)
- Successful login attempts (admin)
- IP address: 127.0.0.1
- User-Agent: Firefox on Linux
- Accurate timestamps for each attempt

```
root@kali: /opt/lampp/htdocs/lab12
Session Actions Edit View Help

(root@kali)-[/opt/lampp/htdocs/lab12]
# cat access.log
[2026-02-15 15:18:23] Login attempt by sadi from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:18:36] Login attempt by him from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:27:49] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:28:36] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:30:38] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:30:58] Login attempt by hero from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:31:07] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:31:47] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:31:59] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS

(root@kali)-[/opt/lampp/htdocs/lab12]
#
```

```
root@kali: /opt/lampp/htdocs/lab12
Session Actions Edit View Help

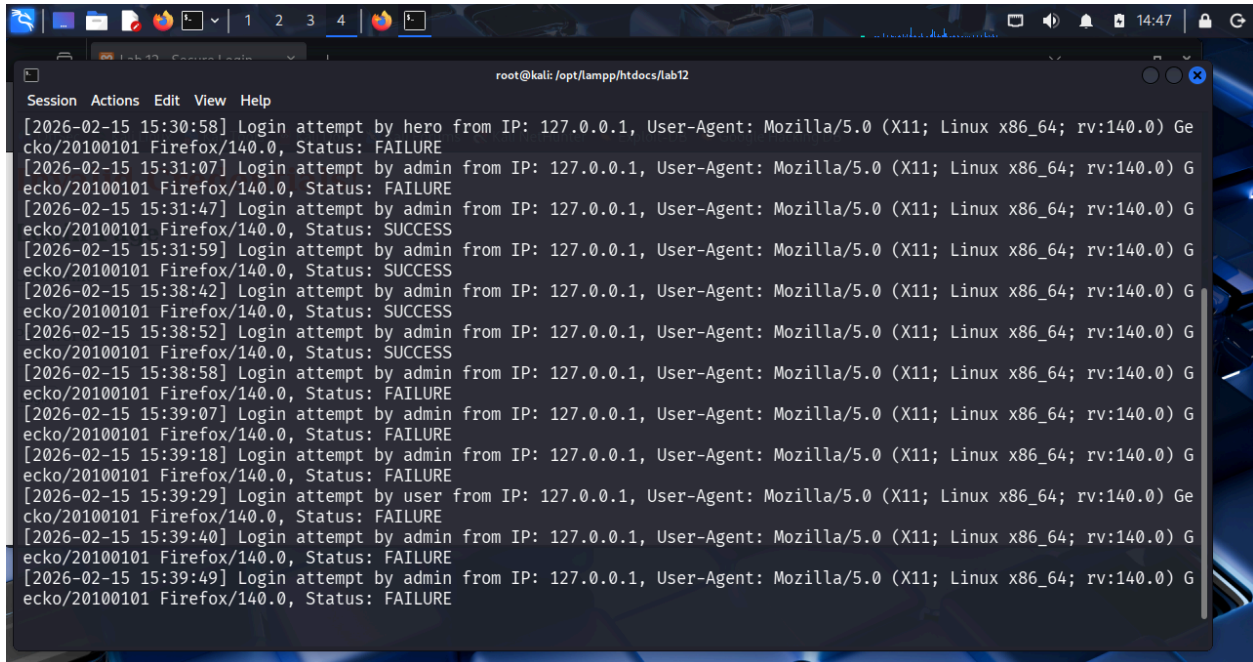
(root@kali)-[/opt/lampp/htdocs/lab12]
# cat access.log
[2026-02-15 15:18:23] Login attempt by sadi from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:18:36] Login attempt by him from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:27:49] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:28:36] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:30:38] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:30:58] Login attempt by hero from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:31:07] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:31:47] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:31:59] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:38:42] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:38:52] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
```

2.3 Detection of Suspicious Activity

From the log:

- Multiple consecutive failures were recorded for admin within a short time frame.
- Repeated failed attempts from the same IP (127.0.0.1) indicate possible brute-force behavior.
- Alternating success and failure attempts show suspicious login testing behavior.

This demonstrates how detailed logs help detect anomalies and unusual activity patterns.



```

root@kali: /opt/lampp/htdocs/lab12
Session Actions Edit View Help
[2026-02-15 15:30:58] Login attempt by hero from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:31:07] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:31:47] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:31:59] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:38:42] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:38:52] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: SUCCESS
[2026-02-15 15:38:58] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:39:07] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:39:18] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:39:29] Login attempt by user from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:39:40] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE
[2026-02-15 15:39:49] Login attempt by admin from IP: 127.0.0.1, User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0, Status: FAILURE

```

2.4 Real-Time Alerting Mechanism

A basic alert system was implemented in login.php:

- If 3 failed login attempts occur from the same IP within 60 seconds,
- An alert entry is written to alert.log.

This simulates real-time monitoring and early attack detection.

3. Reflection Questions

Q1: What information was added to the logs to enhance detection capabilities?

The following information was added:

- Exact timestamp of each login attempt
- Username entered
- IP address of the requester
- User-Agent (browser/system details)
- Login status (SUCCESS or FAILURE)

These details provide full visibility into authentication activities.

Q2: How do these additional details help detect suspicious activities?

- **Timestamps** help identify rapid repeated attempts (brute-force patterns).
- **IP address** helps trace the source of attacks and identify repeated attempts from the same system.
- **User-Agent** helps detect automated tools or unusual clients.
- **Status (SUCCESS/FAILURE)** allows administrators to detect abnormal failure rates.

Together, these details improve monitoring, anomaly detection, and incident response readiness.

4. Conclusion

In this lab, enhanced logging significantly improved visibility into authentication activities. The ability to analyze detailed logs and generate alerts based on suspicious patterns demonstrates how proper logging and monitoring reduce the risk of undetected security breaches.

Real-time alerting ensures that potential attacks can be identified early and addressed before causing serious damage.